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OS Development

3. Programming model of the x86_64 processor architecture

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Preview

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- History
- x86 programming model
- Address translation
- Protection
- Tasks
- Summary

History

- 8086 (1978) - Prime father of the PC processor
- 80286 (1982) - Segmentation based memory protection
- 80386 (1985) - Paging → Memory protection based on pages
- Pentium (1993) - superscalar, 64-bit data bus, MMX, APIC
- Pentium II (1997) - RISC-like micro instructions
- Pentium III (1999) - SSE
- Pentium 4 (2000) - SSE2, Hyperthreading, Intel 64 / EM64T
- Core (2005) - Dual Core
- Core 2 (2006) - Quad Core, 64 bit (x86_64)
- Has-/Broad-well (2013) - AVX2, TSX, FMA3
- Rocket / Tiger Lake (2021) - DL-Boost, Memory encryption
- Alder Lake (2021) - Division in P- and E-cores (max. 8 + 8)

Advanced Vector Extensions (based on SSE)
Transactional Synchronozation Extensions
Fuse-Multiply-Add Instructions

Deep-Learning (DL) Boost
→ AVX extension

P = Performance
E = Efficiency
Hybrid Cores

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8086

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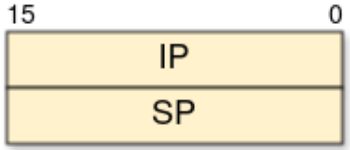
- 16-bit architecture, little endian
- 20-bit address bus → 1 MiB max. physical memory
- Few registers (from todays standpoint)
- 123 instructions (1-4 byte length)
- Segmented memory (64 KiB segments)
- Stil relevant today
  - On multicore systems, other cores startup in real moe

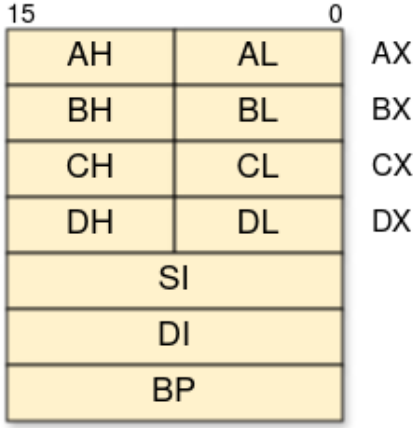
8086: Registers

Instruction and Stack Pointer

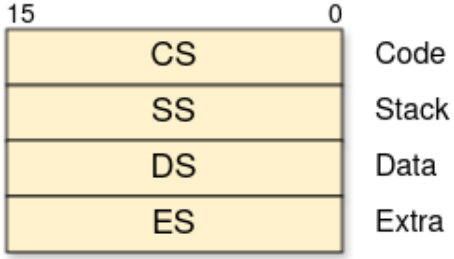


Flags Register

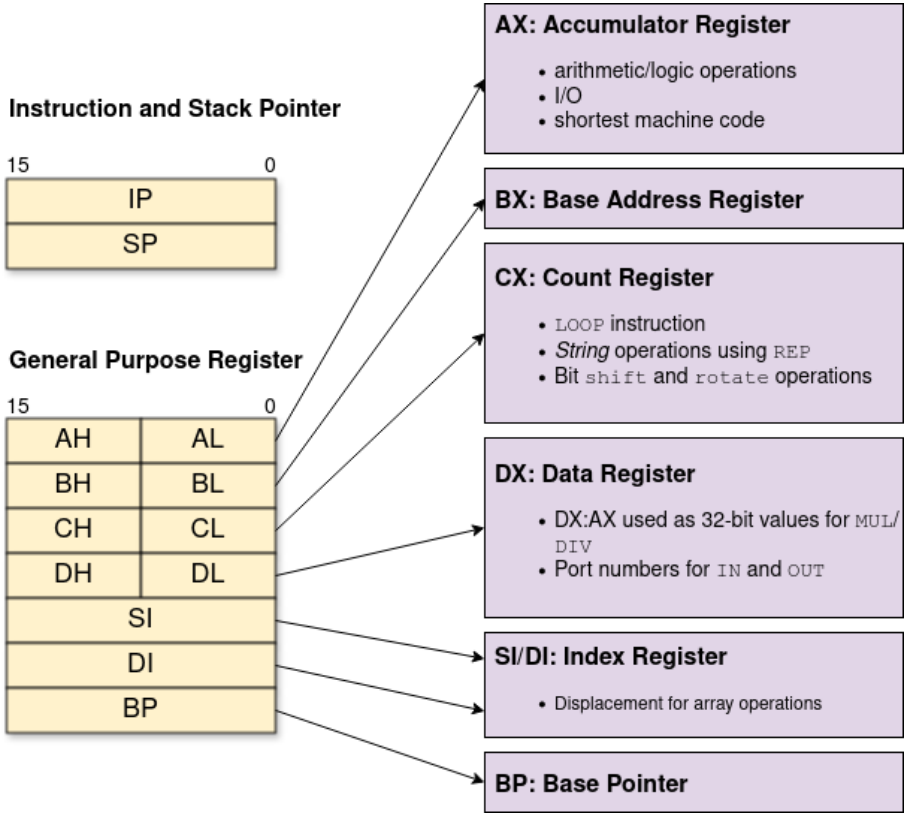
General Purpose Register



Instruction and Stack Pointer

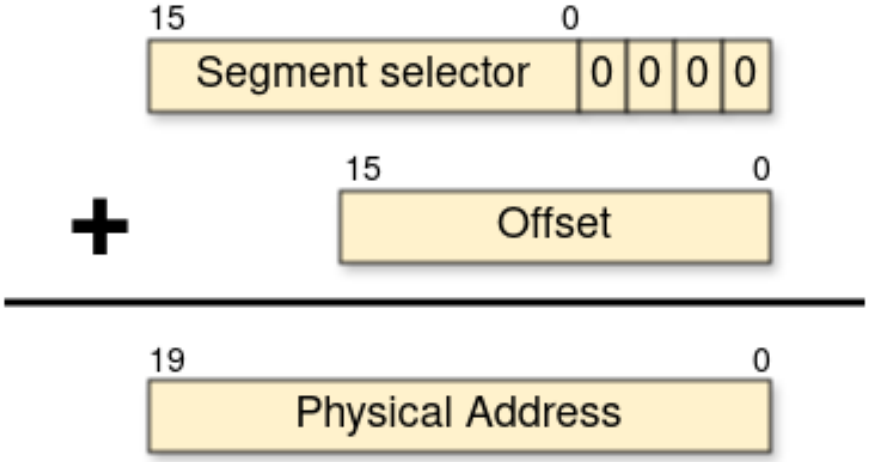


8086: Registers



8086: Segmented Memory

- Logical addresses consist of:
  - Segment selector (determined by a segment register)
  - Offset (usually given by a general purpose register or directly encoded into the instruction)
  - Physical address calculation:



8086: Segmented Memory

8086: I/O-Port vs. Memory Addresses

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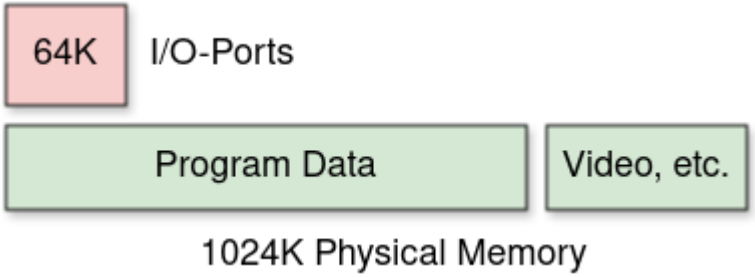
- I/O-port and memory addresses:
  - Differentiation through CPU pin *M/IO*.
  - 16 address lanes for I/O → 16-bit
  - Still supported today (even in Protected Mode and Long Mode)

- Additionally: Memory Mapped I/O (e.g., for video cards)
  - Memory access goes to a device instead of memory



IA-32 – The 32-bit Intel Architecture

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- First IA-32 CPU was the Intel 80386
  - The term *IA-32* was introduced much later
- 32-bit technology: register, data und address bus
  - From Pentium Pro: 64 Bit data und 36 bit address bus
- Complex protection and multitasking support
  - Protected Mode
  - Originally introduced with the 80286 (16-Bit)
- Compatibility
  - With older operating systems through Real Mode
  - With older applications through *Virtual 8086 Mode*
- Segmentation based memory model (mostly used as *Flat Memory*)
- Page based *Memory Management Unit* (MMU)

x86-64: 64-bit CPU Architecture

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- x86-64: Original name of 64 bit extension by AMD
  - Sometimes abbreviated as *x64*
  - Later replaced by *AMD64*
- Intel licensed x86-64 from AMD
- Intel 64 is Intel's own label (for the same architecture)
  - Minimal differences in instruction set compared to AMD64
  - Completely different than the own *IA-64* (Itanium) architechture

x86-64: 64-bit CPU Architecture

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- Additional registers (all 64 bit long)
- Paging must be enabled
- Segmentation strongly restricted
- New feature (among others): *No Execute* protection per page

x86-64: CPU modes

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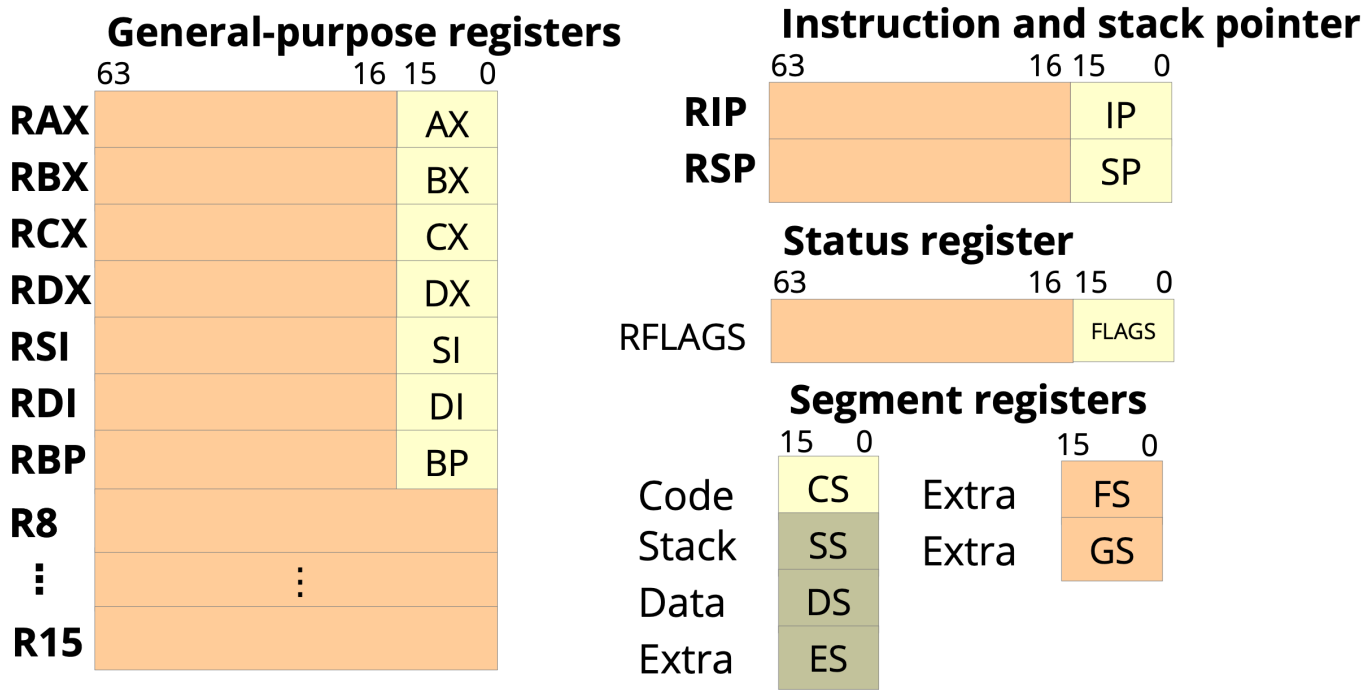
- Real Mode = 16 bit addressing
- Protected Mode = 32 bit addressing (also called IA-32)
- Long Mode = 64 Bit addressing (also called IA-32e)
  - *e* meaning *extension*
  - Used in the Intel manuals
  - To differentiate against IA-64
- (Compatibility Mode)
  - Supporting Protected Mode applications in Long Mode

x86-64: Registers



x86-64: Registers

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Memory-management registers

	15 … 0	63 … 0	31 … 0
TR	TSS sel.	TSS Base Address	TSS Limit
LDTR	LDT sel.	LDT Base Address	LDT Limit
IDTR		IDT Base Address	IDT Limit
GDTR		GDT Base Address	GDT Limit

Details follow ...

Control registers

	63 … 32	31 … 0
CR8		
CR4		
CR3		
CR2		
CR0		

Debug registers

	63 … 32	31 … 0
DR0		
DR1		
DR2		
DR3		
DR6		
DR7		

Model-specific registers (MSRs)

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x86-64: Addressing

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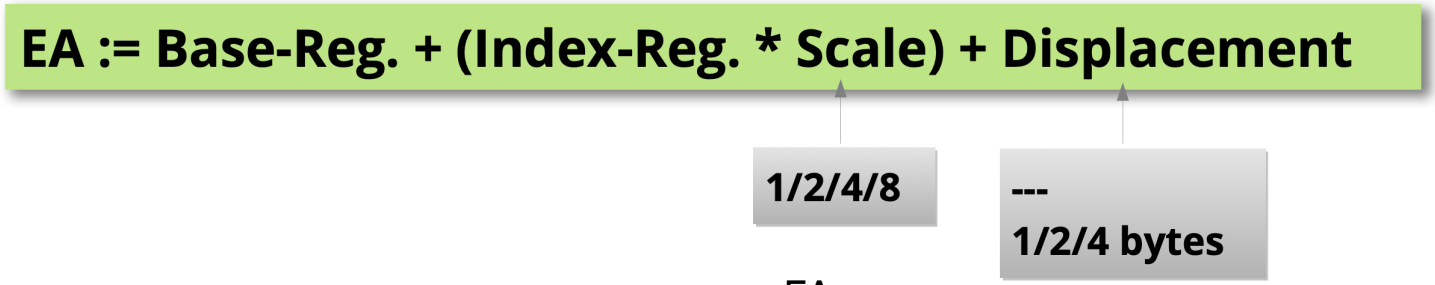
- Effective addresses (EA) are calculated using the following scheme
  - All general purpose registers can be used



- Example: `MOV RAX, array[RSI * 4]` (EA)
  - Lesen aus einem Array mit 4 Byte großen Elementen und RSI als Index
- New in x86-64: Addressing relative to instruction pointer

EA := RIP + Displacement

Protected Mode – Segments

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- Each segment has a start address and a length
- Segment registers store *segment selectors* = Index into the GDT/LDT
  - LDT was rarely used by operating systems
- A program (in execution) consists of multiple memory segments
  - At least CODE, DATA and STACK
- *Linear address* is segment start address + effective address
  - Effective address = Value in a general purpose register
  - Linear address = physical address, if the paging unit is turned off

Protected Mode – Segments

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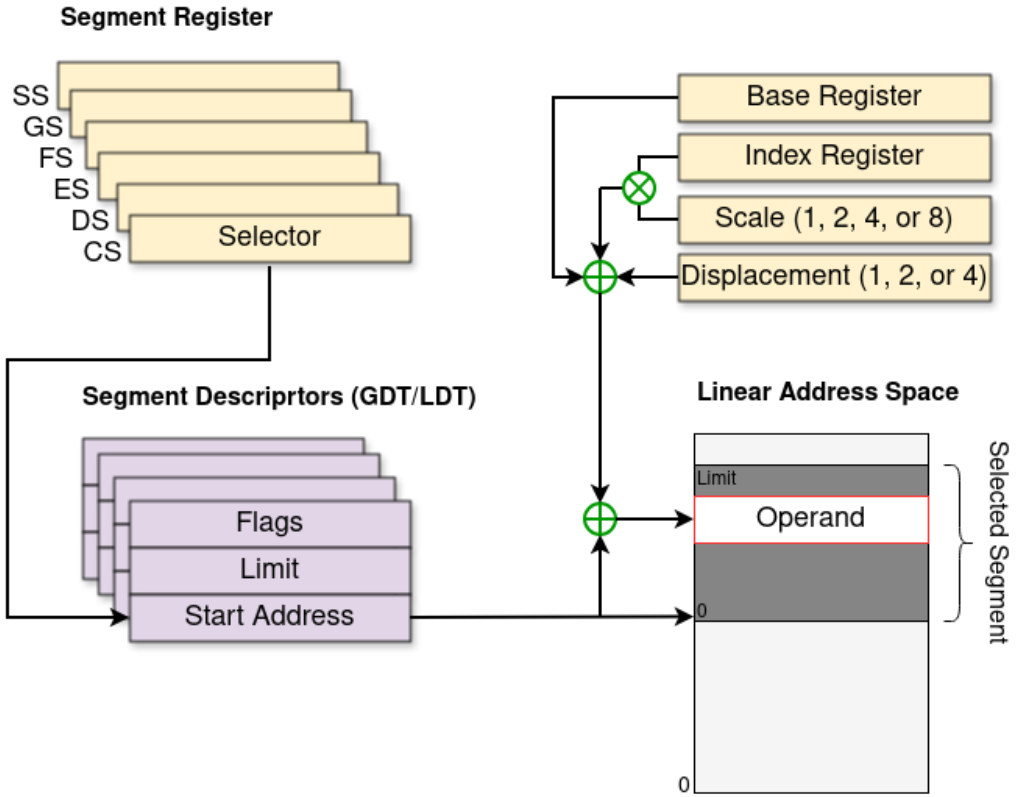
- Flat memory:
  - Practically all modern 32-bit operating systems set all segments to start address 0
  - This way, segments do not influence address calculation
  - Furthermore, only four GDT entry are used (CODE + DATA, each for *User* and *Kernel Mode*); LDT remains completely unused

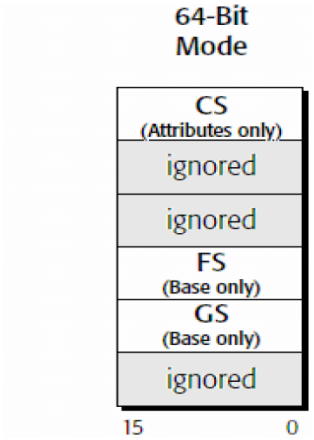
- Segmentation cannot be turned off (because of protection), but Paging is not required in IA-32 Protected Mode

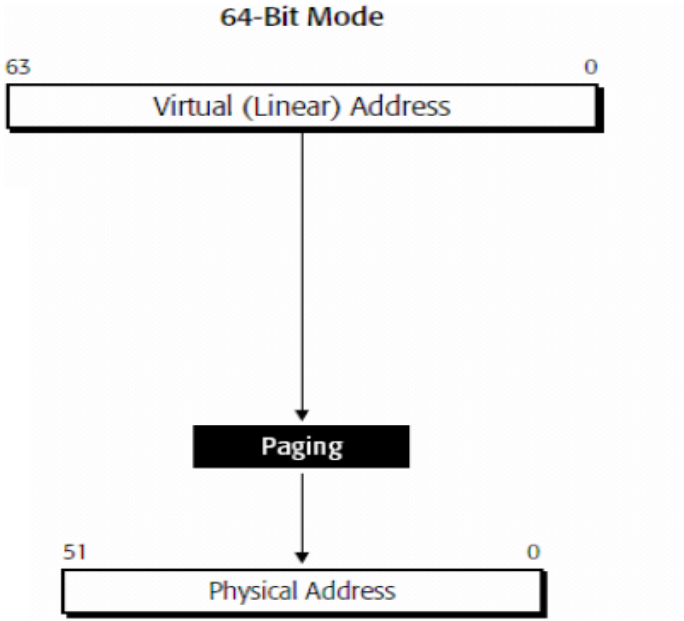
Protected Mode – Segments



Long Mode: Segmentation

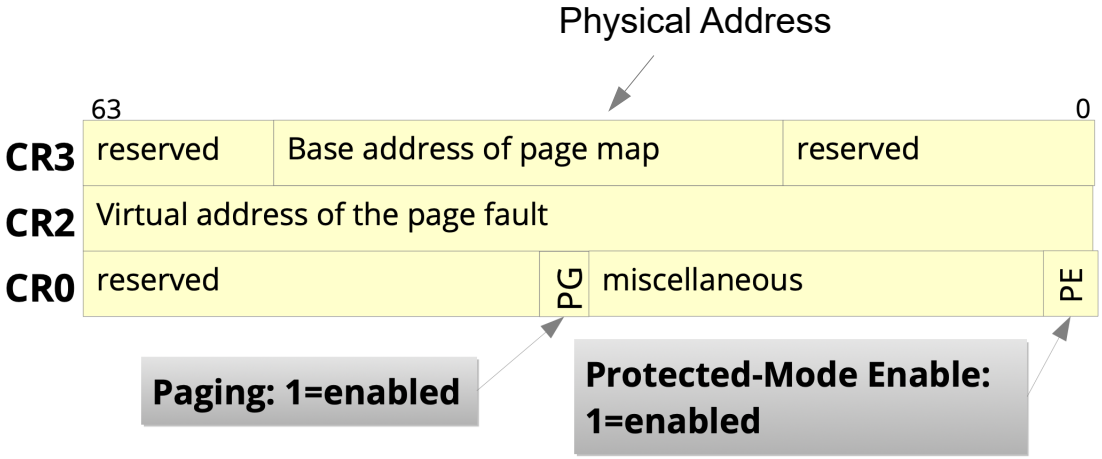
- 16-bit Selectors
- Start addresses are ignored in almost all segments, except FS and GS
- No limit check; essentially only the ring number in CS remains relevant (= CPL)





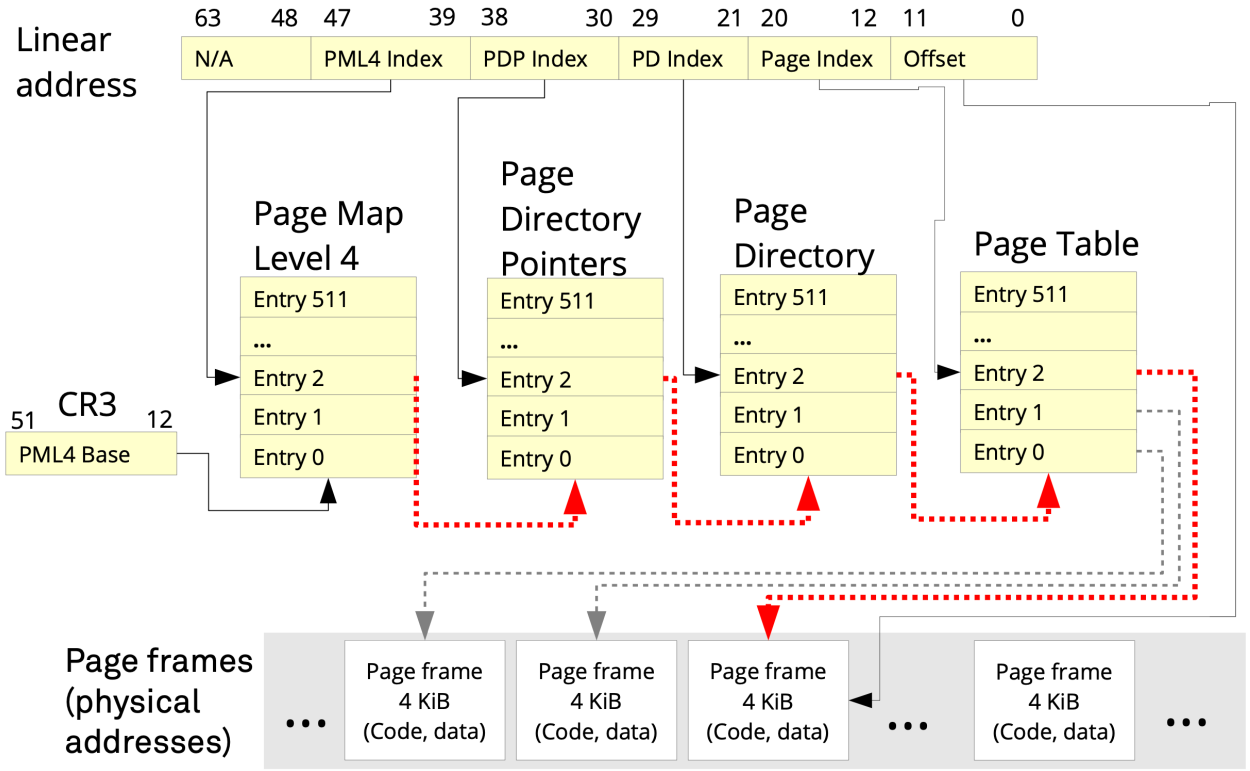
Long Mode: Paging

- Paging is required in Long Mode
- The basic paging configuration is handled by the CRx **C**ontrol **R**egisters



Long Mode: Page Tables

- Four or five levels of page tables



Long Mode: TLB

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- Problem: Paging slows down memory accesses, since the page tables need to be accessed each time
- Solution: *Translation Lookaside Buffer* (TLB):
  - Fully associative cache
  - Tag: consists of page table indices
  - Data: Page Frame address
  - Size (80386): 32 entries
- With normal applications, the TlB reaches a hit rate of ~98%
- Remarks:
  - Writing the CR3 register invalidates the TLB
    - Not anymore since Intel Westmere (2010) → TLB Tag has additional 12-bit *Process Context ID* (PCID)
  - When protection/access bits in a page table entry are modifired, the corresponding TLB entry must be deleted (INVLPG instruction)

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Protection

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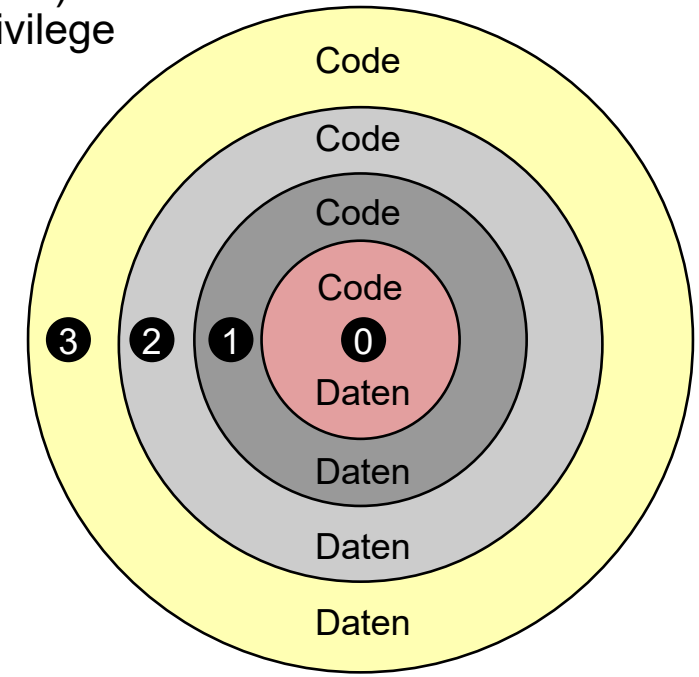
- Goals:
  - Protection against unauthorized memory accesses and code execution
  - Protection against kernel crashes or crashes of other processes

- Requirements: Coda and Data...
  - are classified in regard to their trustability
  - and this classification is enforce by the operating system, with the help of the process

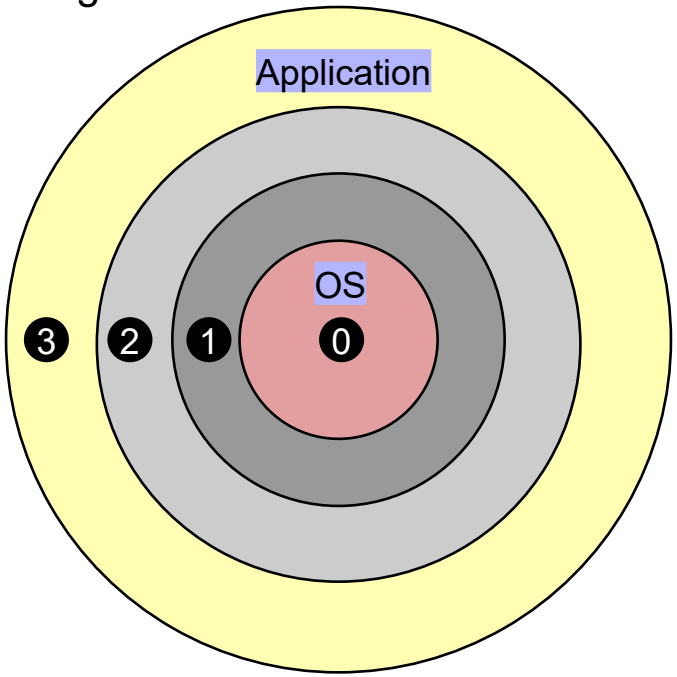
Protection Rings

- Each segment descriptor (Code/Data) has two bits, determining which privilege level the segment is assigned to
  - Up to four privilege levels possible
  - In practice, only two are used



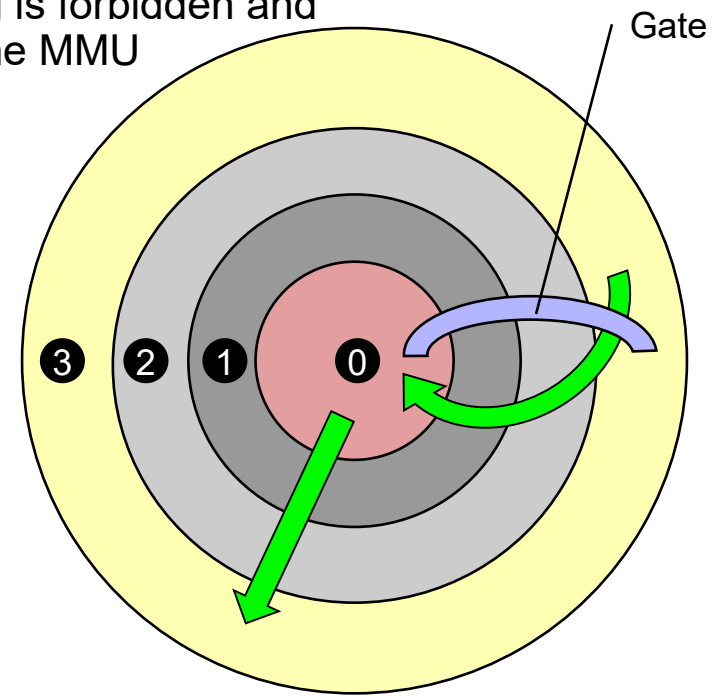
Protection Rings

- Smaller ring numbers grant more privileges
- Ring 3 is meant for applications
  - = User Mode
- Ring 0 is meant for the OS
  - = Kernel/Supervisor Mode



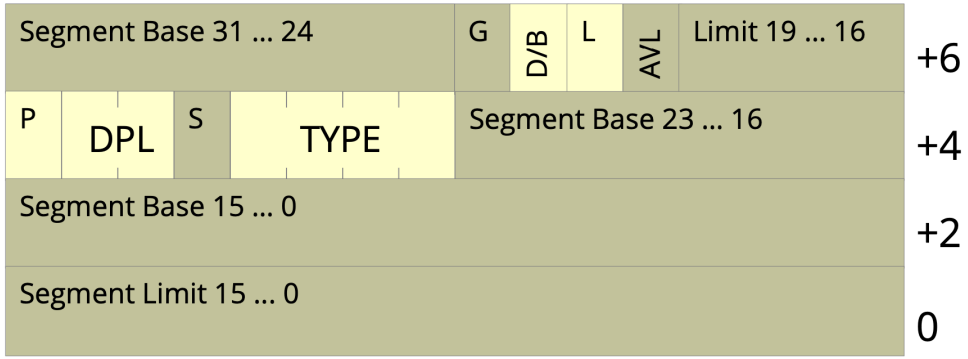
Protection Rings and Gates

- Accessing memory of an inner ring is forbidden and such accesses are prohibited by the MMU
  - General Protection Fault
- Code of an inner ring can only be called via **Gates**
  - Allow direct calls from ring 3 to ring 0
- Memory accesses and calls to outer rings are always allowed



Segment Descriptors

- Are entries in the Global Descriptor Table (GDT) and allow code protection
  - (IA-32: additional protection via limit (= end of segment))
  - 8 byte per entry (Exception: system segments use 16 byte in x86_64)



TYPE
Code/Data
Read/Write
Non-/Conforming

System Segments (Call-Gate, Task State Segment)

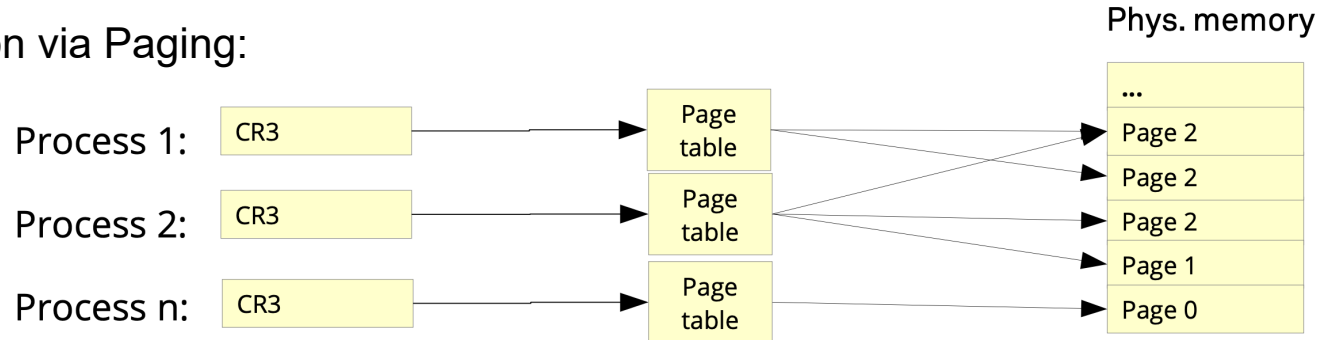
P	- Present Bit
DPL	- Descriptor Privilege Level
S	- System Segment

G	- Granularity
D/B	- 16/32 Bit Seg.
L	- Long Mode aktiv

Flat Memory Model

- Most 32-bit PC operating systems have not used segmentation
  - 32-bit effective address = linear address
- In Long Mode (64 bit) the segment base address is ignored by the processor

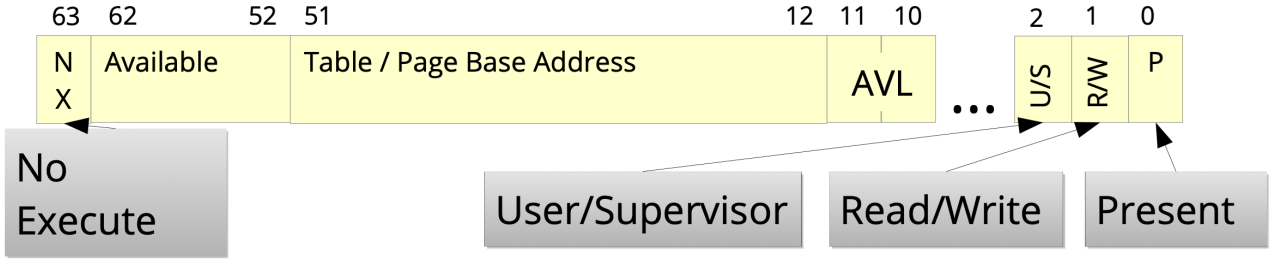
- Protection via Paging:



- Each process only sees its own *virtual address space*
  - Page level access control (see next slides)

Page Table Entry

- Flags are settable on all levels of the page table hierarchy
- Protection bits:
  - R/W = Read/Write → Protection agains write accesses
  - NX = No eXecute → Do not execute code inside data
  - U/S = User/Supervisor → Access allowed for all or only in ring 0



  - Available = AVL = These bits are free to use by the operating system

x86_64: Summary - Protection

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- Modern 64-bit operating systems only need the followin GDT entries
  - One data segment descriptor
  - Two code segment descriptors (one with DPL = 0 and one with DPL = 3)

- Protection is enforced via Paging
  - Protection bits (NX, U/S, R/W)
  - Page Tables:
    - Which pages are present/accessible?

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IA-32: HW Support for Multitasking

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- State of a task is stored in a *Task State Segment* (TSS)
  - Stores all registers
  - Stack pointer for each ring
  - CR3 register
  - I/O permission bitmap (see next slides)
- Current TSS determined by *Task Register* (TR) of a CPU core
  - Each core has its own TR
- Task switch with hardware support via a *Task Gate*
  - Not used in most operating systems and instead implemented in software

Long Mode: Multitasking

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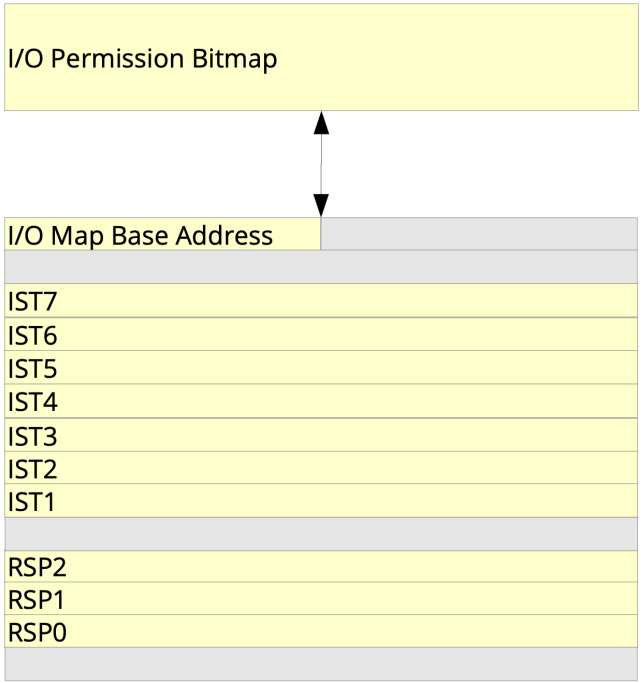
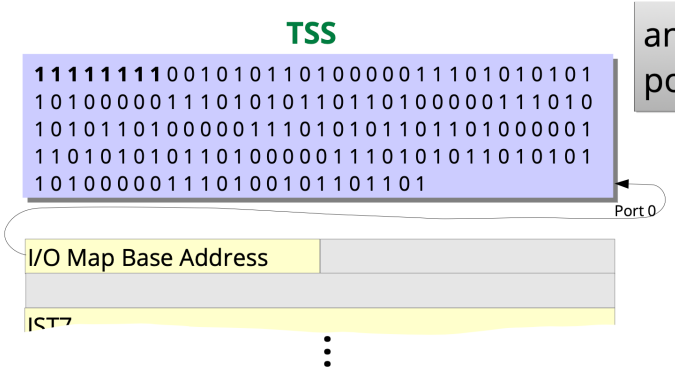
- No HW support for task switching
- TSS existent, but only stores:
  - Stack pointer per ring
  - IST = *Interrupt Stack Table*
  - I/O permission bitmap (see next slide)
- IST allows separate stacks for interrupt handlers
  - IST number can be specified in entries inside the Interrupt-Descriptor Table (IDT)



Long Mode: I/O Permission Bitmap

- Access protection for I/O-ports (only little relevance nowadays)
  - I/O privilege level (IOPL) represented by two bits inside the RFLAGS register
  - Port access is granted if CPL ≤ IOPL
  - Otherwise, the I/O permission bitmap is used
    - One bit for each port; if a bit is set, access is denied
- Access protection for memory mapped I/O via page tables

Bitmap ends with the TSS segment's end, ports with higher numbers must not be accessed.



an 1 prevents port access

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- x86-64 architecture is highly complex
  - Virtual memory with four or five level paging
  - Page based memory protection
  - I/O-port access protection per task
  - Can execute 16-bit code in IA-32 Protected Mode or 32-bit code in Long Mode
- Rarely used features were removed
  - Segmentation (mostly)
  - Hardware support for task switching
- Still backwards compatible
  - Processor and cores still start up in Real Mode

More Features

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- Virtualization
  - Virtual 8086 Mode
    - 16-bit application
  - Intel-VT, AMD-V
    - HW support for VMware, VirtualBox, etc.
    - Hypervisor runs in ring -1
- AVX: Advanced Vector Extensions
- TSX: Transactional Synchronozation Extensions
- Hybrid Cores